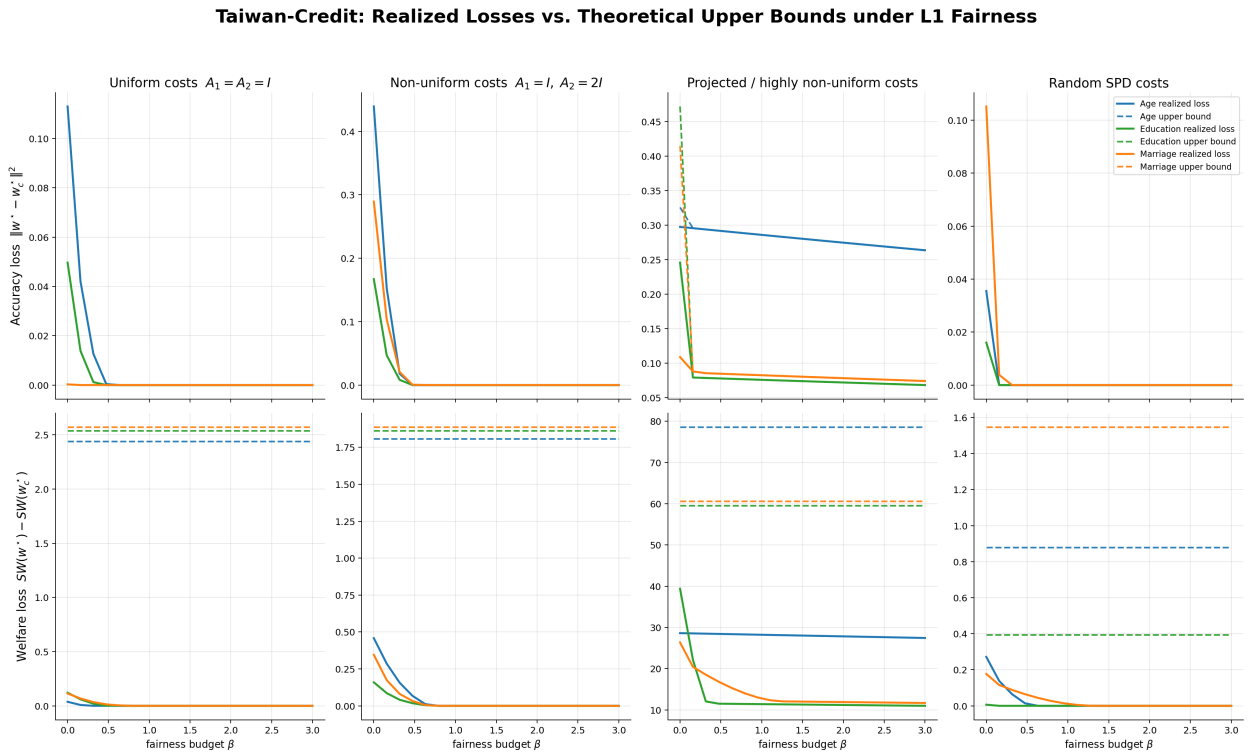

Figure 1: Comparison of the principal's realized accuracy loss ($\|\mathbf{w}^\star - \mathbf{w}_c^\star\|^2$) and social welfare loss ($\mathrm{SW}(\mathbf{w}^\star) - \mathrm{SW}(\mathbf{w}_c^\star)$) against our theoretical upper bounds (Table 1, Property 3.1) across varying fairness budgets β. **Experimental Setup**: We simulate the principal-agent interaction using the real-world Taiwan-Credit dataset. To accurately model how agents exert effort, we construct the contribution matrix C using the empirical causal graph derived by [Sello and Taryn, 2025] (https://www.mdpi.com/2227-9091/14/2/38). This structural equation model captures the realistic direct and indirect causal pathways between financial features. **Group Formation & Subspaces**: To study the effect of underlying data geometry on the tightness of the fairness constraint, we define three separate demographic partitionings: 1. Age: Younger (≤ 25 years) vs. Older (> 25 years). 2. Education: Higher Education (graduate school & university) vs. Lower Education (high school & others). 3. Marriage: Married vs. Not-married. For each demographic subgroup g, we perform Singular Value Decomposition (SVD) on their historical data to extract the top $k = 5$ principal components, defining their natural variance subspace Π_g.